INCH-POUND

MIL-T-27/3B
23 September 2009
SUPERSEDING
MIL-T-27/3A
6 December 1976

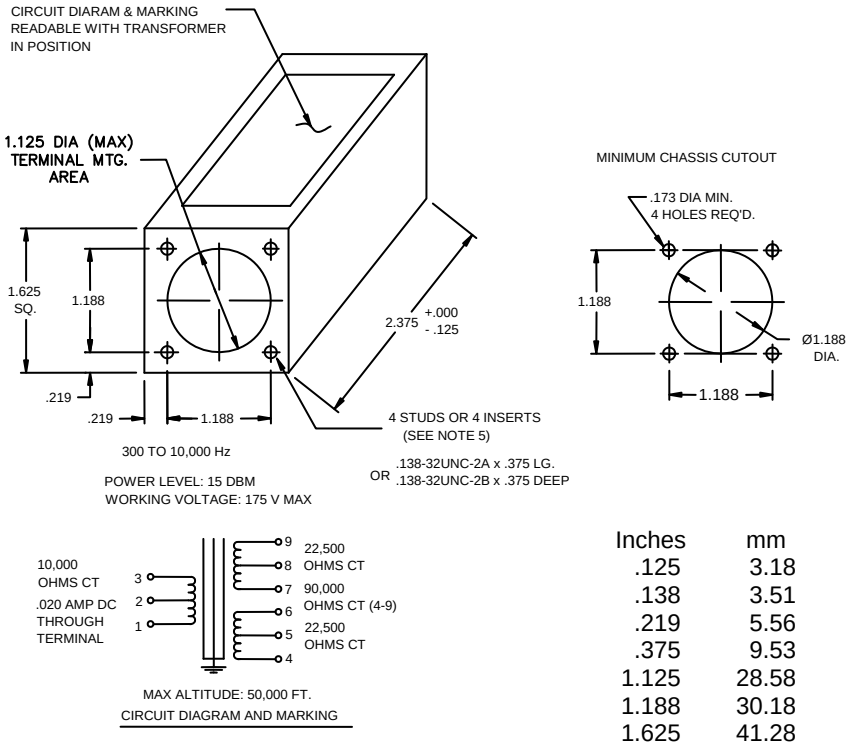
MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY

Inactive for new design
after 25 August 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.



Inches	mm
.125	3.18
.138	3.51
.219	5.56
.375	9.53
1.125	28.58
1.188	30.18
1.625	41.28
2.375	60.33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on overall case dimensions is +.000 (0.00 mm) -.063 (-1.60 mm).
4. Tolerance on mounting dimensions is +.016 inch (0.41 mm).
5. Inserts not recommended for new design.
6. Terminal arrangement optional within mounting area indicated.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS: (When numbers in parentheses, such as (1-3) are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, (4-9), the windings are connected in series, terminals 6 and 7 are connected.)

Electrical ratings:

Primary power level: +15 dBm, maximum; -8 dBm, minimum.

Working voltage
(1-3) - 175 V, maximum.
(4-6) - 175 V, maximum.
(7-9) - 175 V, maximum.

Primary current: .020 ampere dc through terminal 2; .001 ampere dc, maximum unbalance.

Primary impedance (1-3): 10,000 ohms ct.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Terminals: Solder terminals, No 20 AWG.

Height: .312 inch, +.000, -.156 (7.92 mm +0.00 -3.96).

Weight: .72 pound, maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Operating temperature: 105°C, maximum.

Dielectric withstanding voltage: MIL-STD-202, method 301; test voltage at sea level, 500 V rms. At reduced barometric pressure, 300 V rms.

Electrical characteristics:

Harmonic distortion: Not more than 1 percent with 20 V, 300 Hz across (1-3) and .020 ampere dc through terminal 2.

No load (centertap voltage unbalance only): With 20 volts, 1,000 hertz across (1-3) and .020 dc through terminal 2:
Unbalance (4-5) and (5-6): 1 percent, maximum.
Unbalance (7-8) and (8-9): 1 percent, maximum.
Unbalance (4-6) and (7-9): 1 percent, maximum.
Unbalance (1-2) and (2-3): 1 percent, maximum.

Secondary load impedance: (4-6) – 22,500 ohms ct.
(7-9) – 22,500 ohms ct ± 10 percent.
(4-9) – 90,000 ohms ct.

Primary impedance (1-3): With approximately 1 V, 1,000 Hz across (1-3) and .020 ampere dc through terminal 2, and 90,000 ohms ct across (4-9), the primary impedance shall be 10,000 ohms ct ±10 percent across (1-3).

Frequency response: Range – 300 to 10,000 Hz at rated power, flat within ±2 dB; reference frequency = 1,000 Hz; test voltage 1 Vrms (1-3).

Insertion loss: 1.0 dB maximum, with 20 V, input at 1,000 Hz across (1-3).

Resonance: Second resonant frequency = 20,000 Hz, minimum.

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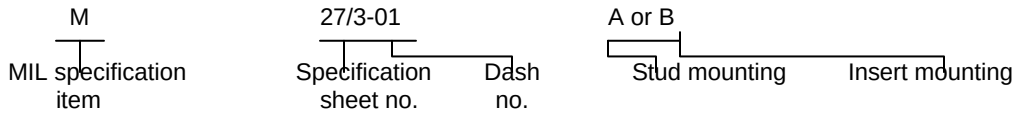
Polarity: Additive, with terminals 3 and 4, and 6 and 7 connected.

Temperature rise: 40˚C maximum, above an ambient of 65˚C, with rated power at 300 Hz.

Vibration (High frequency): MIL-STD-202, method 204, test condition B.

Shock (specified pulse): MIL-STD-202, method 213, test condition I.

Part or Identifying Number (PIN):



(For cross reference of MS PIN to military PIN, see table I.)

TABLE I. Cross reference.

Mounting	MS9000		M27/3	
	Dash no.	Type designation	Dash no.	Type designation
Studs	-A	TF4RX13AJ001	-01A	TF4R21AJ
Inserts	-B	TF4RX13AJ001	-01B	TF4R21AJ

NOTE: Qualification approval to MIL-T-27/3 shall be sufficient to grant qualification approval to MIL-T-27/4 through MIL-T-27/6.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-015)

Review activities:
Army – AR, CR4
Navy – AS, MC
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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